

The background of the slide is a photograph of a traffic jam on a multi-lane road at night. The scene is dimly lit, with the primary light sources being the headlights and taillights of the cars, and the lights of a traffic signal on the right side of the frame. The traffic signal has three circular lenses; the top one is red, the middle one is green, and the bottom one is yellow. The cars are packed closely together, stretching into the distance. In the background, there are silhouettes of buildings and trees under a dark sky. The overall color palette is dominated by dark blues, greys, and the warm yellows and reds of the artificial lights.

SMART TRAFFIC SIGNAL OPTIMIZATION

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The Problem

1

Rigid by design

Cycle time (90-180s) is split equally across every lane, no matter how many vehicles are waiting.

2

Blind to demand

The system cannot tell a packed lane from an empty one - so empty roads get green while queues build elsewhere.

3

Fails under stress

Accidents or surges can't be handled; officers must step in and direct traffic manually.

4

Costly to the country

Estimated to cost the Sri Lankan economy over Rs. 1 billion every day in fuel, time and productivity.

A National-Scale Cost

What idle engines and blocked intersections actually cost Sri Lanka

Rs. 1B+

lost to the economy
every single day

Millions

of person-hours
wasted idling in
queues each week

Beyond the direct cost: wasted fuel drains foreign exchange reserves, and idling engines drive up air pollution and greenhouse-gas emissions in crowded city centers.

The Proposed Solution

1

Sense

Cameras on the signal poles count cars, buses, three-wheelers and motorcycles waiting in every lane, in real time.



2

Rank

A Max-Heap instantly identifies which lane has the longest queue and needs help the most.



3

Allocate

A Greedy algorithm hands out green time in proportion to real demand - busiest lane gets the most.



4

Protect

A mandatory 10-second minimum green time is guaranteed to every lane, so none is ever starved.

Existing System: Fixed-Timer Control

How it works

Total cycle time is divided directly by the number of lanes (n). Every lane gets an identical share, cycle after cycle - regardless of how many vehicles are actually present.

Decision complexity

$O(1)$ to split the cycle time

- $O(n)$ to assign it to all lanes

Cheap to compute - but completely blind to real queue conditions.

Baseline Simulation Output

North	avg 206.20	max 412
South	avg 76.12	max 142
East	avg 1.45	max 8
West	avg 1.17	max 8

Overall average queue
length

71.24

Proposed Algorithm: Greedy + Max-Heap

1

Reserve the safety minimum

4 lanes $\times$ 10s minimum green = 40s reserved. Remaining pool = 90 - 40 = 50 seconds.

2

Heapify the queues — $O(n)$

Vehicle counts are arranged into a Max-Heap; the busiest lane rises to the top instantly.

3

Extract-Max — $O(n \log n)$

The system repeatedly pops the busiest lane, and the heap reshuffles to reveal the next.

4

Distribute proportionally

If North holds 60% of total demand, it receives 60% of the remaining 50s on top of its 10s minimum.

Developed System: Simulation Results

Greedy + Max-Heap Output

North	avg 15.21	max 34
South	avg 10.73	max 32
East	avg 1.83	max 9
West	avg 1.62	max 8

Overall average queue **7.35**

89.7%

reduction in overall average queue length

(71.24 → 7.35 vehicles)

13x

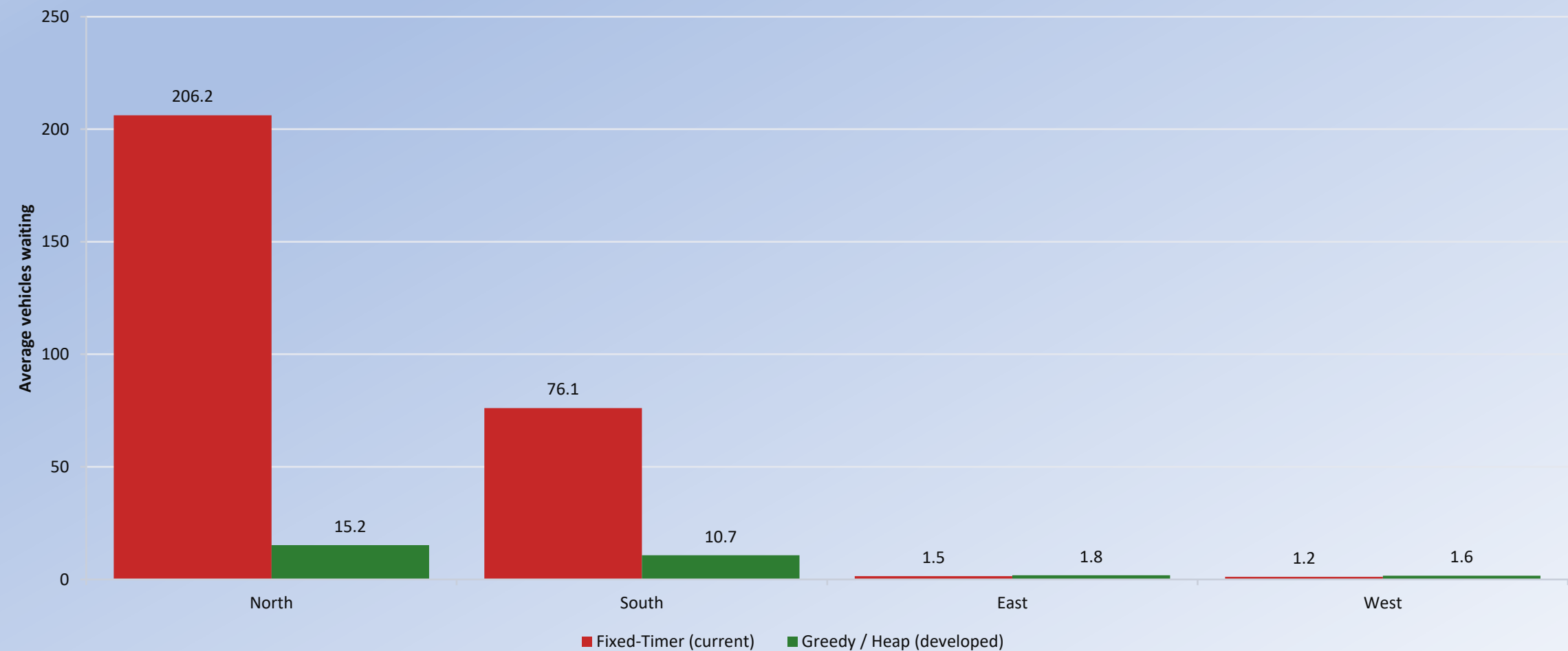
improvement on the busiest lane,

North (206.20 → 15.21 vehicles)

*East and West stay low and stable in both systems
- The greedy method fixes the main roads without sacrificing minor lanes.*

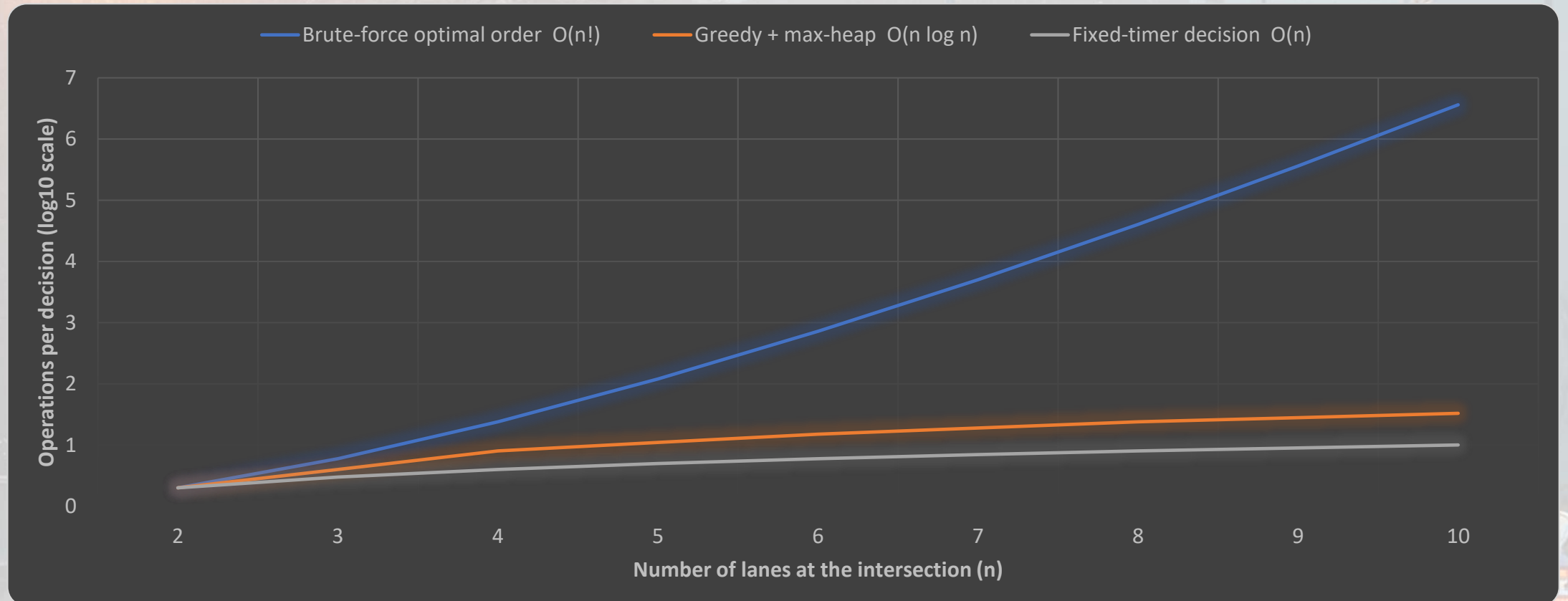
Average Queue Length per Lane

Fixed-Timer (current) vs. Greedy / Max-Heap (developed)



Complexity Comparison

Per-cycle decision cost as the number of lanes (n) grows



At n = 10 lanes: brute force needs ~3.6 million operations; the greedy heap needs only 33.

Conclusion

The current fixed-time system costs Sri Lanka over Rs. 1 billion a day - equal green time for every lane regardless of real demand.

7.35

vehicles average queue
(down from 71.24)

89.7%

overall reduction in traffic
queues

13×

faster clearing on the
busiest lane (North)

By sensing real demand instead of guessing, the Greedy + Max-Heap system eliminates unnecessary jams, saves fuel and money, and cuts emissions - all with a decision cost only marginally higher than today's fixed timer.

The background of the image is a dark blue, semi-transparent overlay on a photograph. The photograph shows an aerial view of a multi-lane roundabout with several cars circulating. To the right of the roundabout, a vertical traffic light is visible, with its red, green, and yellow lenses clearly shown. The text "Thank you" is centered over the roundabout area.

Thank you